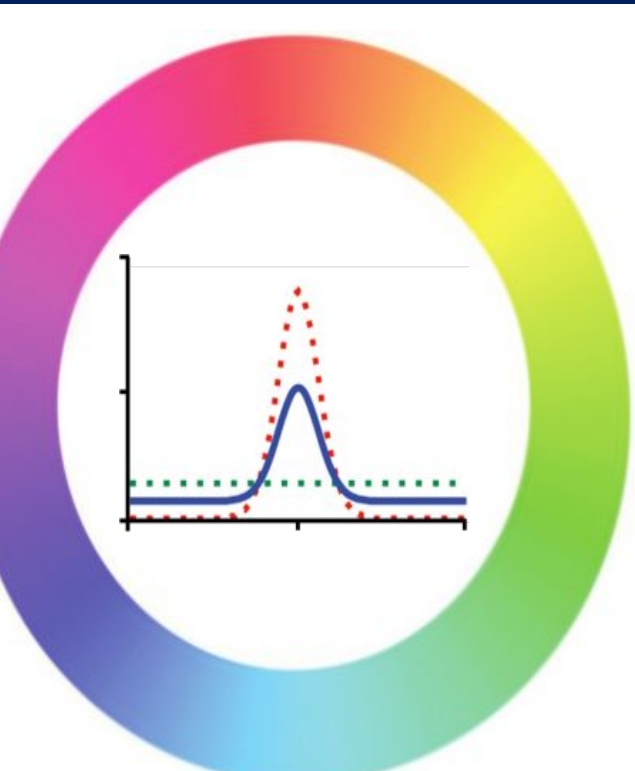




The Dissociation between Capacity and Precision in Visual Working Memory for Motion

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INTRODUCTION

General Background

Working Memory: the ability of retaining information shortly

- Capacity : probability of successful recall
- Precision : quality of memory representation

WM precision for colors¹, shapes¹, and objects, like Pokémon², is dissociable from WM capacity (probability of recall).

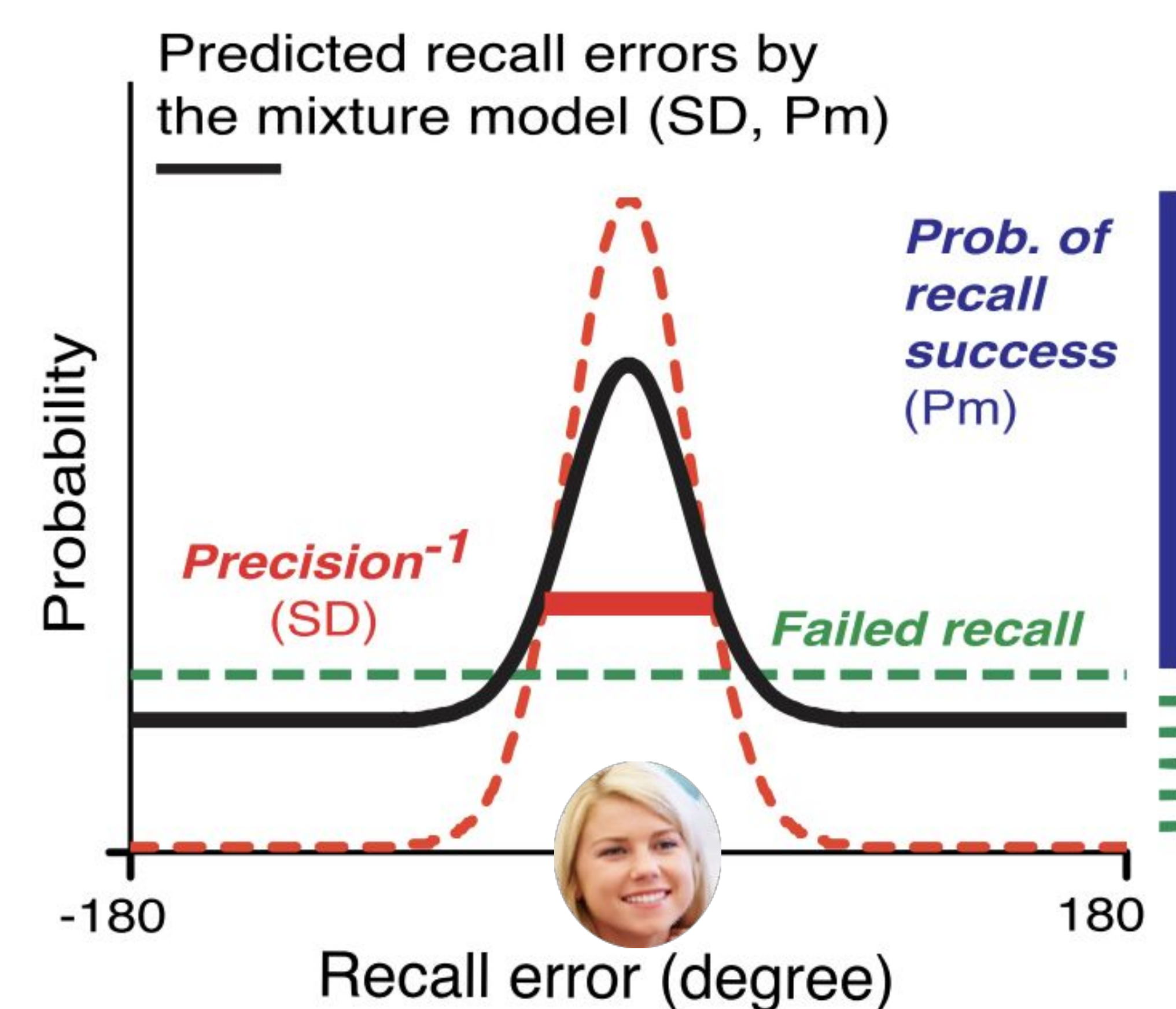
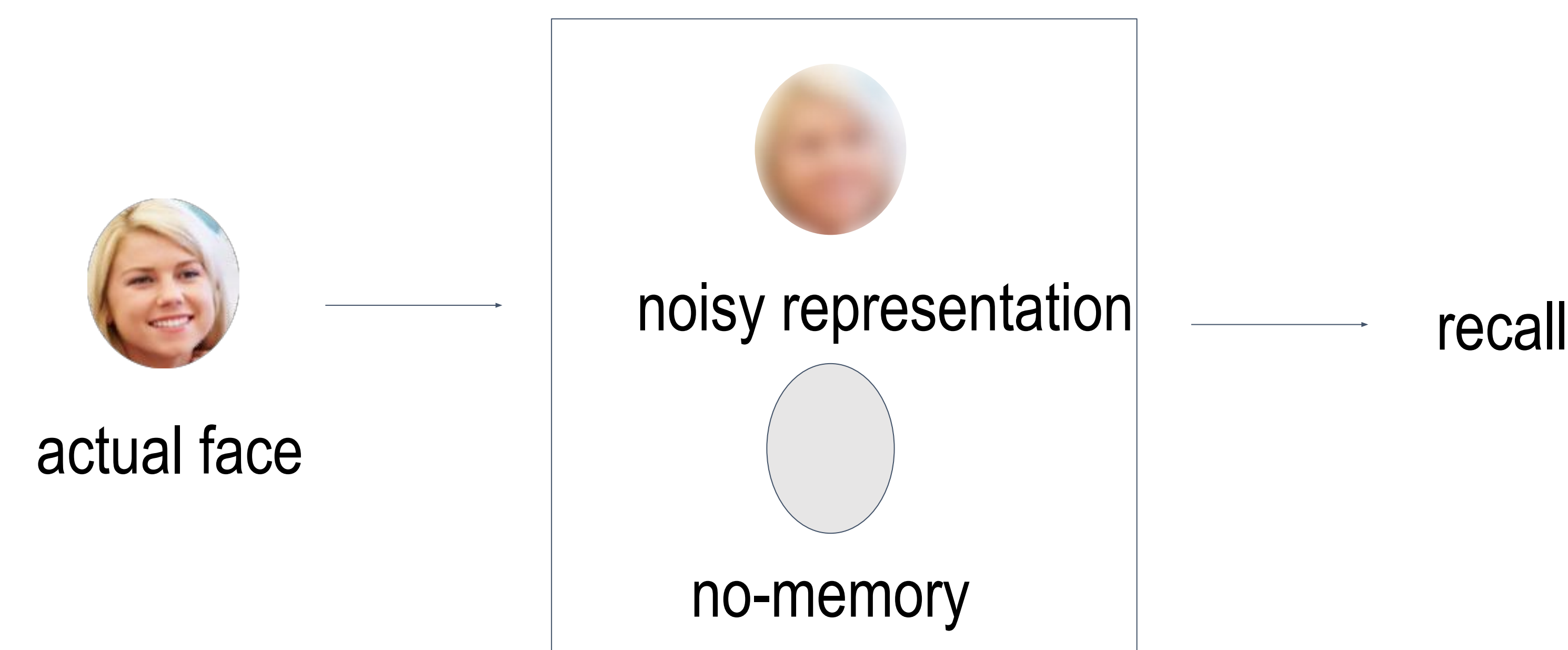
- Unclear whether this dissociation can be applied to a dynamic information such as motion signal

Hypothesis

The capacity and precision of working memory for motion are dissociable

Mixture Model (Zhang and Luck, 2008)

Memory recall is driven by a mixture of noisy representations (precision) and the probability of successful memory (capacity)



- Recall performance (black) reflects a mixture of:

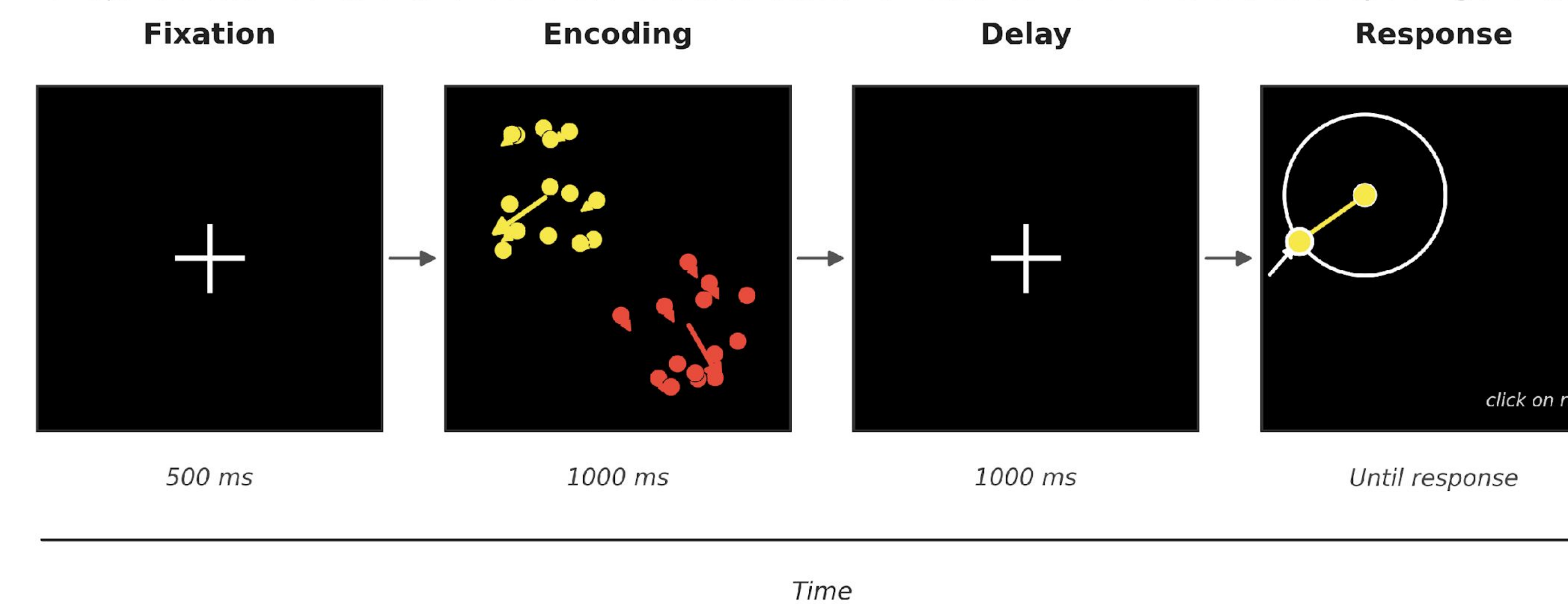
- Memory-based representation (red): responses centered on the target
- Capacity-limited guessing (green): random responses unrelated to the input

METHODS AND RESULTS

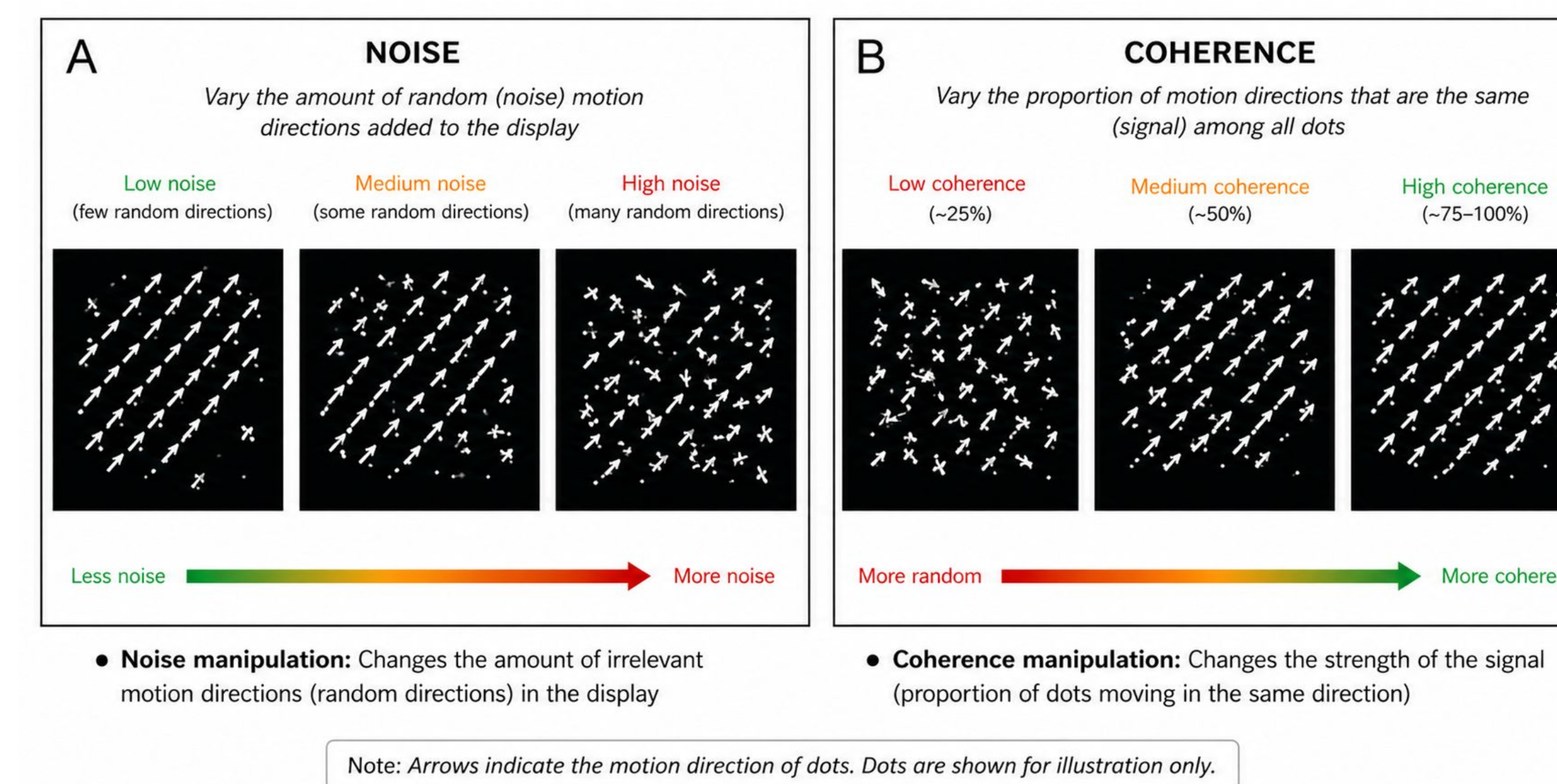
Stimuli and Task (RDM)

Motion-direction working memory task

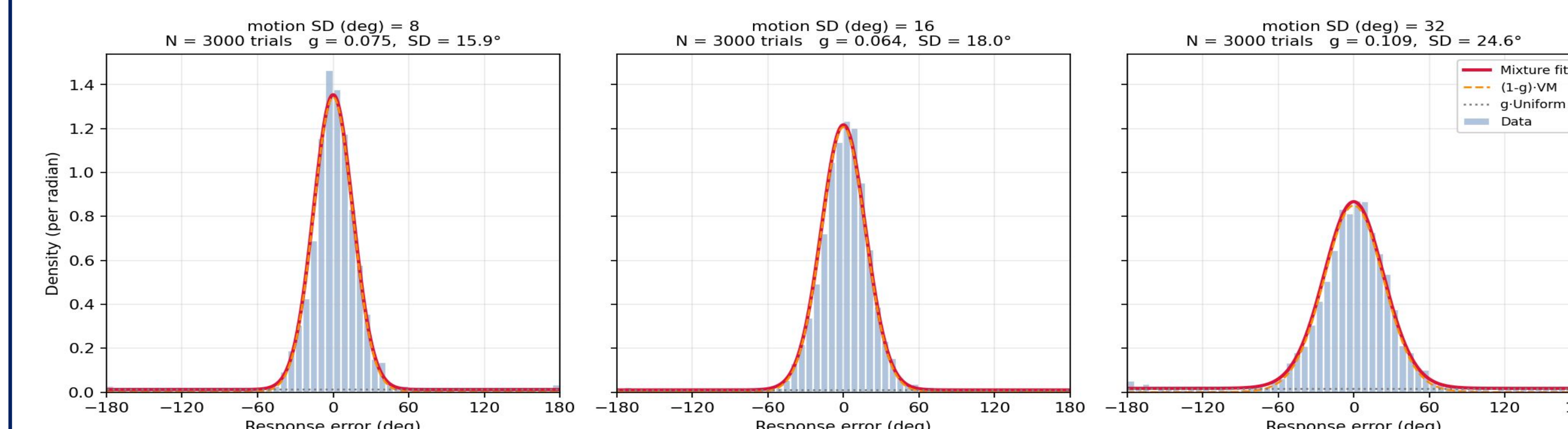
Participants encode the direction (and noise level) of colored dot clouds, then reconstruct the remembered direction by clicking on a response ring.



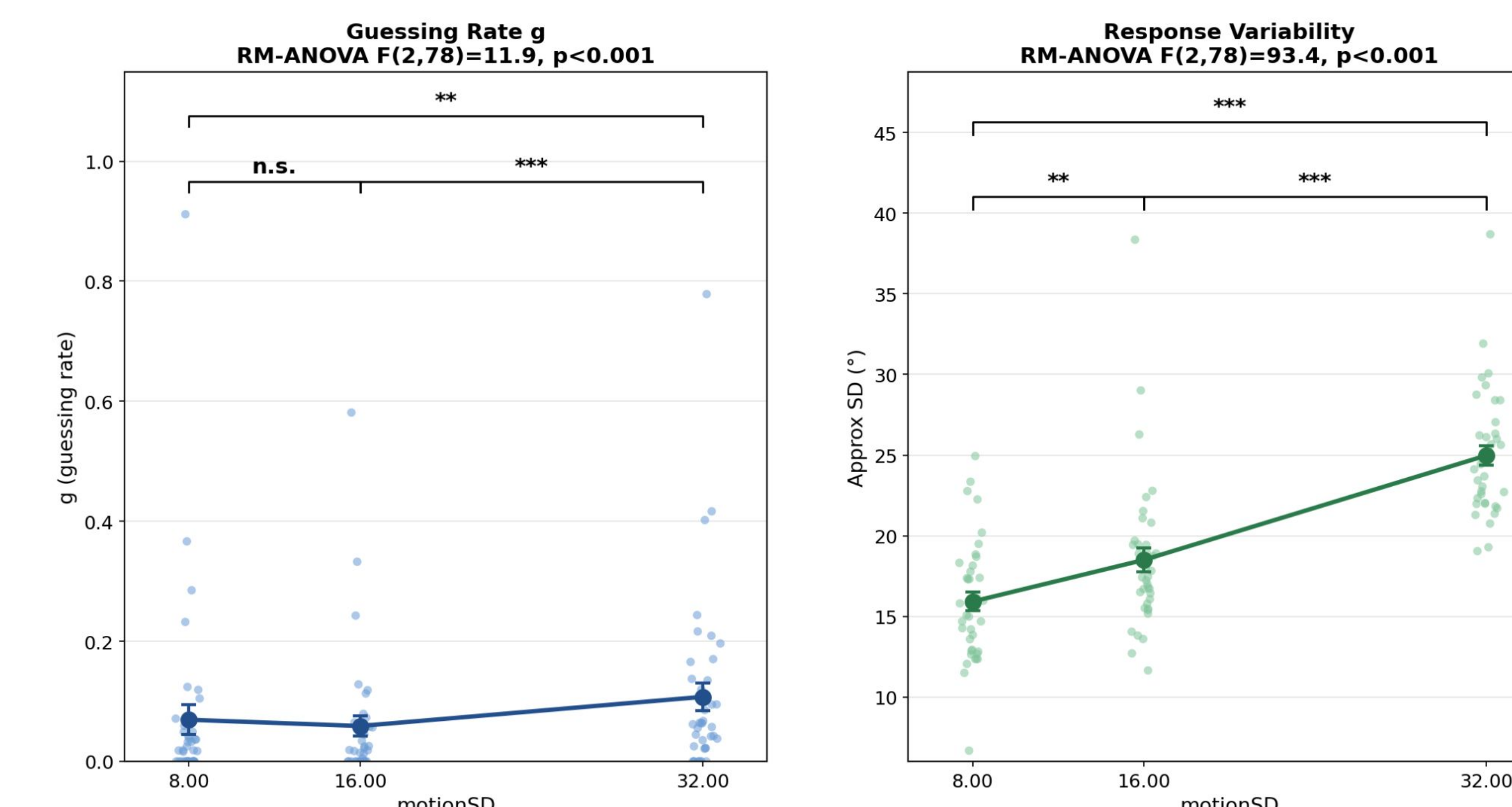
Experimental Manipulation



Experiment 1, motion direction noise will degrade WM precision



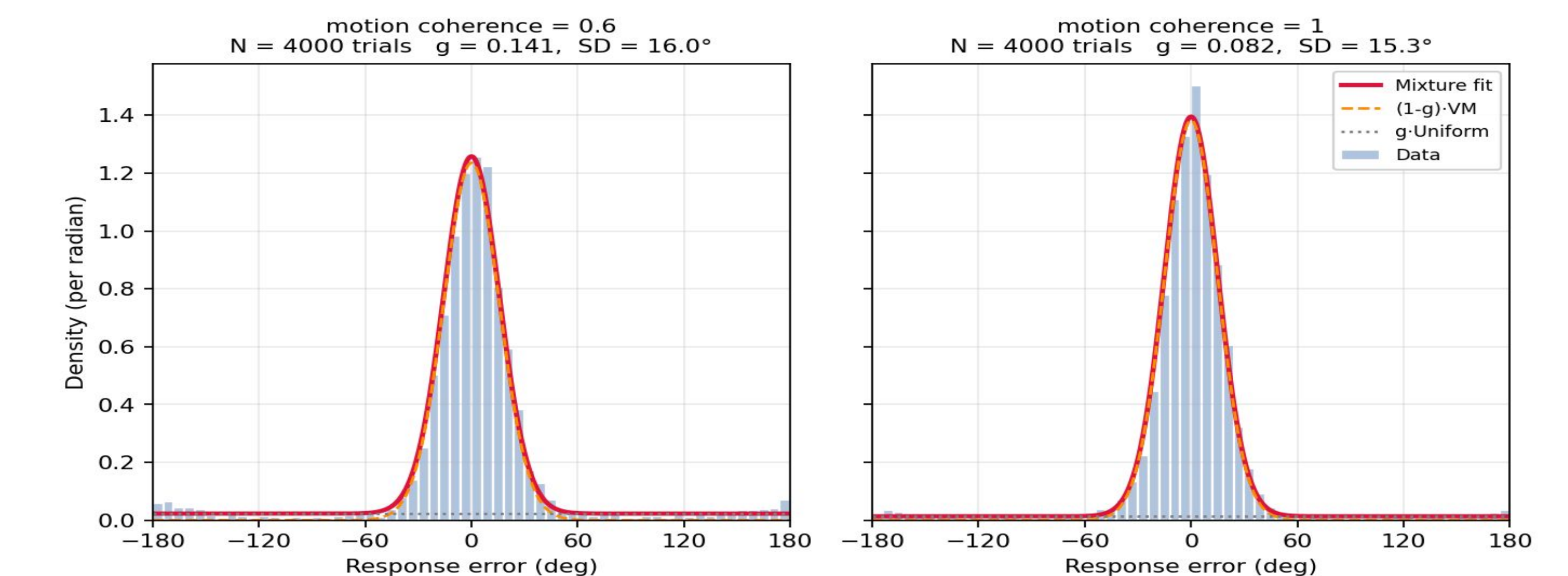
Experiment 1 — Individual Data — N=40 Included Subjects (motionSD; mean±SEM; Holm-corrected paired t)



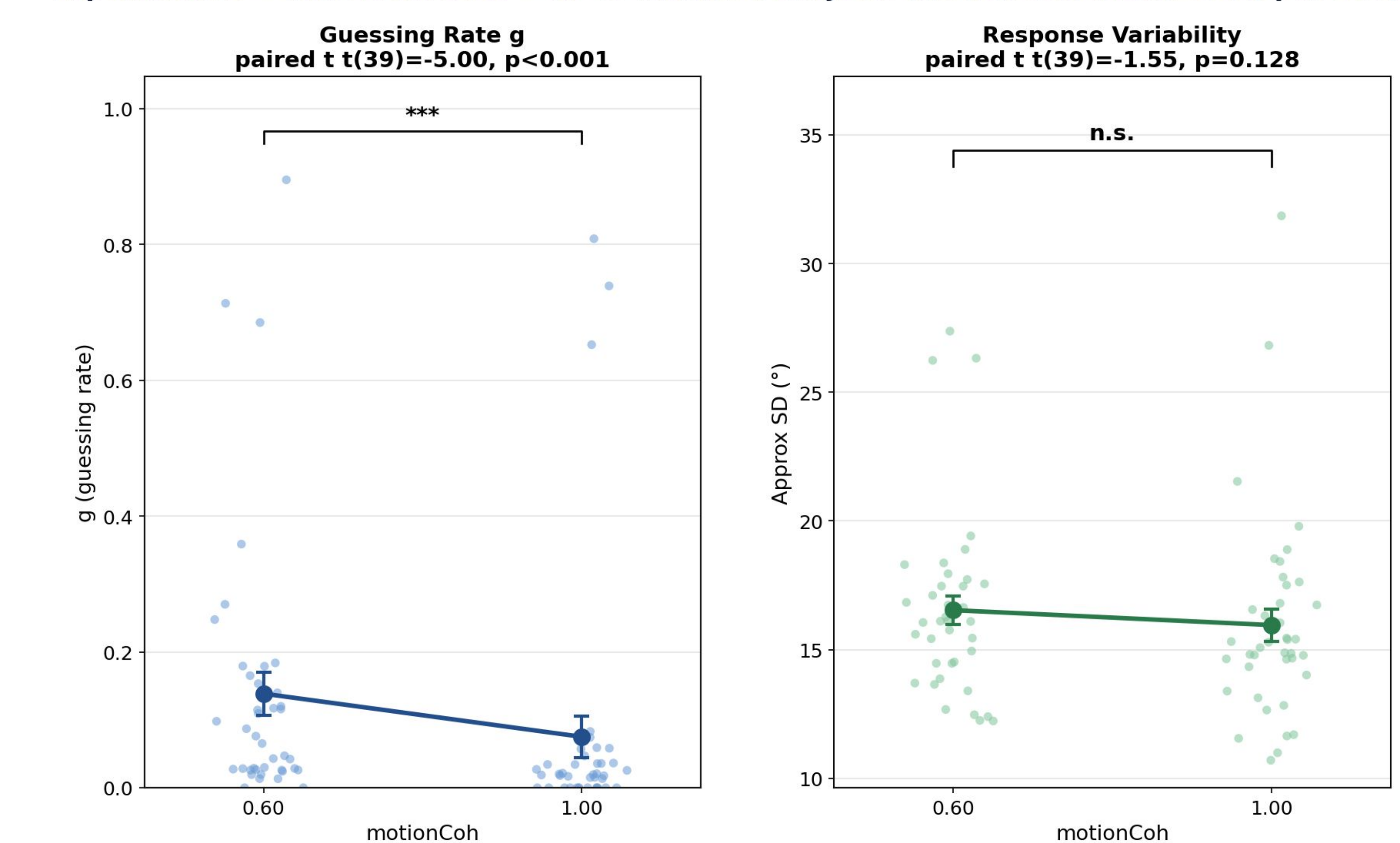
- Increasing motion noise (sd 8 -> sd 16) led to high recall variability with small changes in recall success.
 - Extremely large motion noise (sd 32) worsen WM precision and capacity
- WM precision, not capacity, was reduced by motion direction noise

Experiment 2, motion coherence will increase WM capacity

Experiment 2 — Zhang & Luck (2008) mixture model, fit per condition (aggregated across 40 subjects; pilots excluded)



Experiment 2 — Individual Data — N=40 Included Subjects (motionCoh; mean±SEM; paired t)



- Increasing motion coherence selectively reduces guessing with little impact on recall variability
- WM capacity, not precision, was increased by signal-to-ratio in motion information

Discussion and Conclusions

- WM precision and capacity for motion were dissociable, Consistent with previous work^{3,4}
 - Variability and signal-to-noise ratio of motion differentially affected recall variability and success
- Future directions:
 - Dissociate the neural substrates (where; e.g., hippocampus) and mechanism (how; e.g., pattern separation) underlying WM capacity and precision for motion
 - Use fMRI/EEG to dissociate neural signatures of capacity and precision

References

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- Xie, W., & Zhang, W. (2017). Familiarity increases the number of remembered Pokémon in visual short-term memory. *Memory and Cognition*, 45(4), 677–689. <https://doi.org/10.3758/s13421-016-0679-7>
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- Bae, G. Y., & Luck, S. J. (2022). Perception of opposite-direction motion in random dot kinematograms. *Visual Cognition*, 30(4), 289–303. <https://doi.org/10.1080/13506285.2022.2052216>